SIMULATIONS UPDATE
ANNA SOFIA MORO

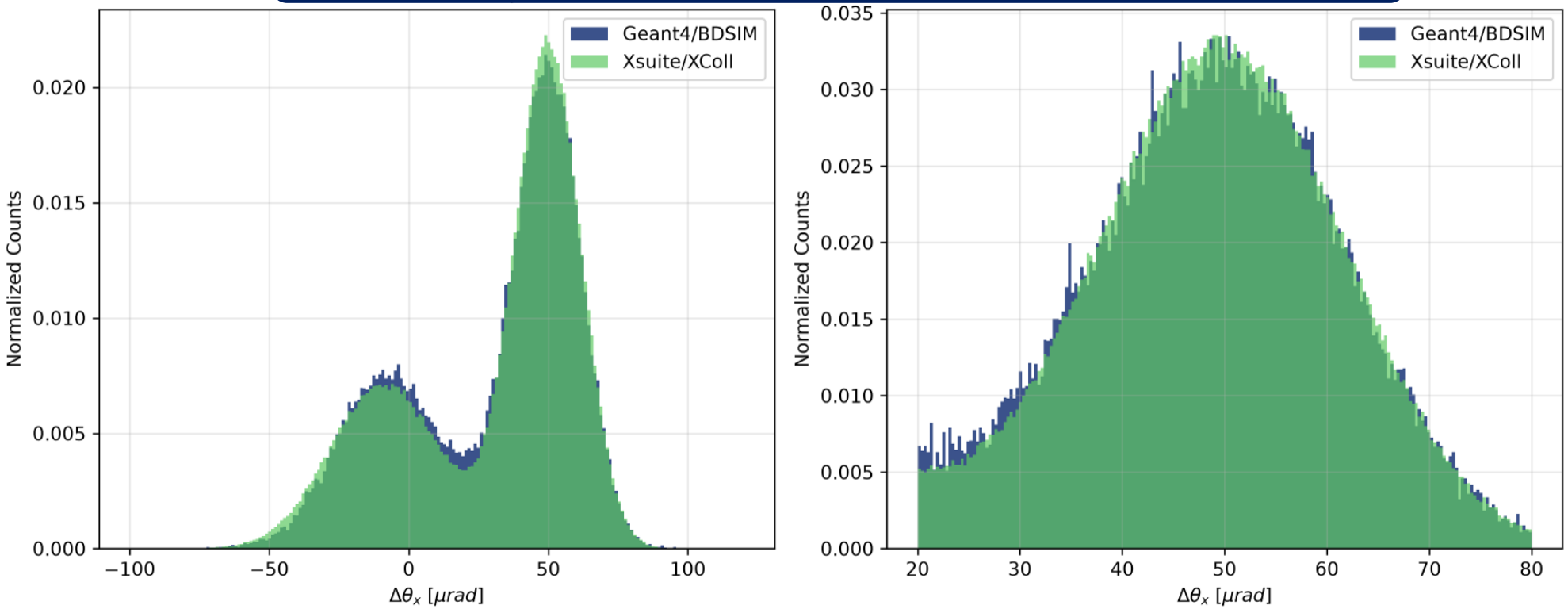
SIMULATIONS

	BDSIM (GEANT4)	XCOLL (XSUITE)	DATA
SHORT CRYSTAL			$\varepsilon_{ch} = (61.9 \pm 0.5)\%$
LONG CRYSTAL			$\varepsilon_{ch} = (19.6 \pm 0.2)\%$

- Same beam distribution as real data
- Manually added detector resolution $\approx 9\ \mu\text{rad}$
- Same analysis

SHORT CRYSTAL - COMPARISON

$\varepsilon_{ch} = (64.8 \pm 0.2)\%$ $\varepsilon_{ch} = (62.6 \pm 0.5)\%$

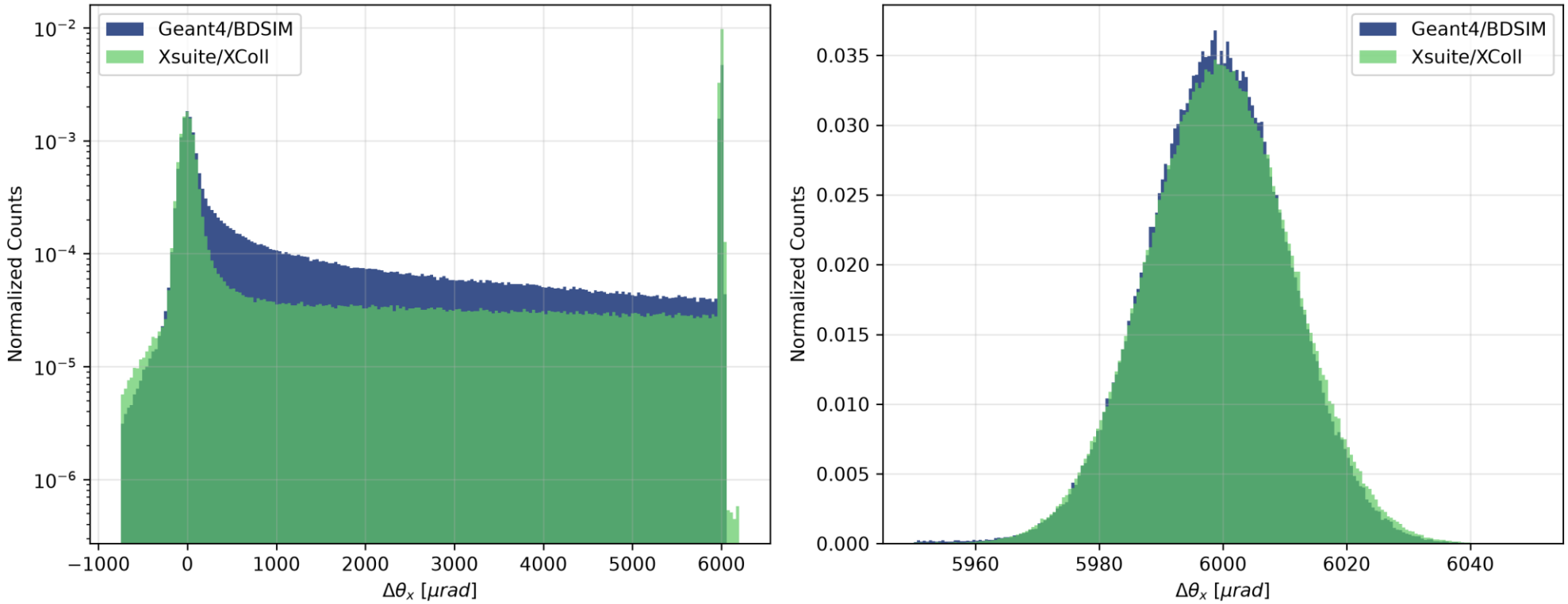


$\theta_b = 49.9 \pm 0.1\ \mu\text{rad} \quad -\sigma = 11.7 \pm 0.1\ \mu\text{rad}$

$\theta_b = 49.8 \pm 0.1\ \mu\text{rad} \quad -\sigma = 11.8 \pm 0.1\ \mu\text{rad}$

LONG CRYSTAL - COMPARISON

$\varepsilon_{ch} = (41.2 \pm 0.1)\%$ $\varepsilon_{ch} = (20.2 \pm 0.1)\%$



$\theta_b = 5999.5 \pm 0.1\ \mu\text{rad} \quad -\sigma = 11.5 \pm 0.1\ \mu\text{rad}$

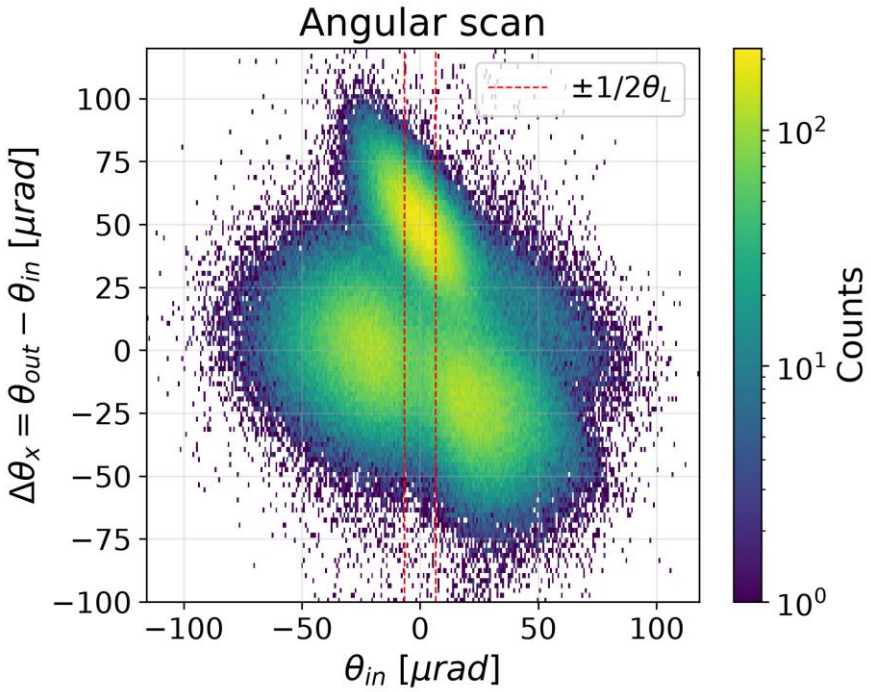
$\theta_b = 5999.3 \pm 0.1\ \mu\text{rad} \quad -\sigma = 11.0 \pm 0.1\ \mu\text{rad}$

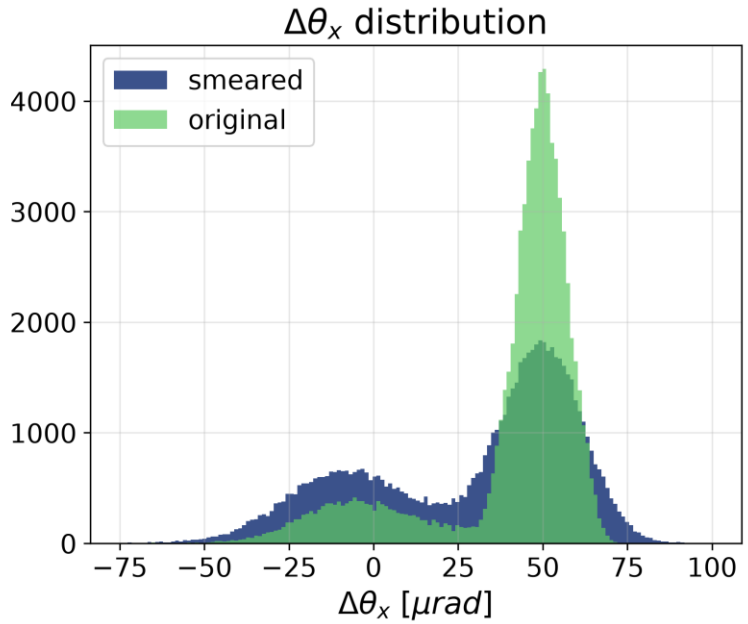
RESULTS

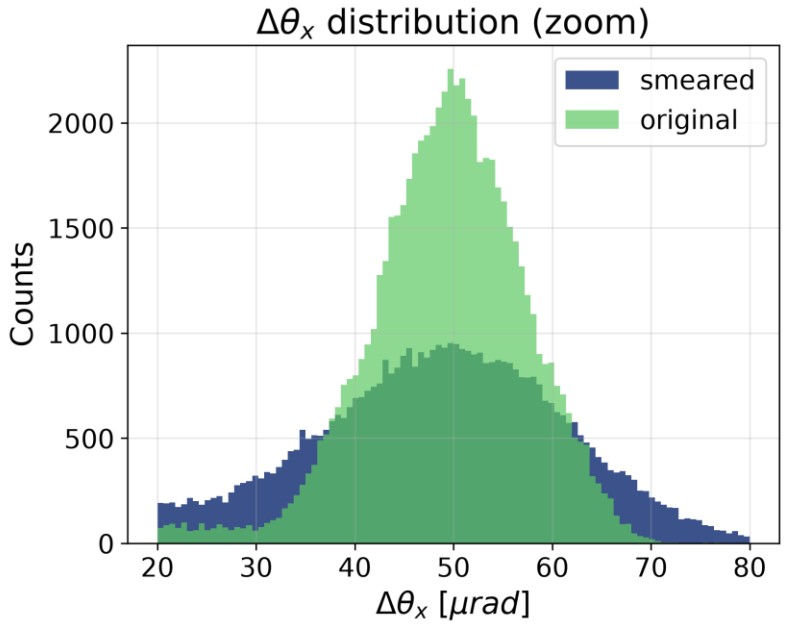
	BDSIM (GEANT4)	XCOLL (XSUITE)	DATA
SHORT CRYSTAL	$\varepsilon_{ch} = (62.8 \pm 0.6)\%$	$\varepsilon_{ch} = (64.8 \pm 0.2)\%$	$\varepsilon_{ch} = (61.9 \pm 0.5)\%$
LONG CRYSTAL	$\varepsilon_{ch} = (20.2 \pm 0.1)\%$	$\varepsilon_{ch} = (41.2 \pm 0.1)\%$	$\varepsilon_{ch} = (19.6 \pm 0.2)\%$

BACKUP

SHORT CRYSTAL: BDSIM



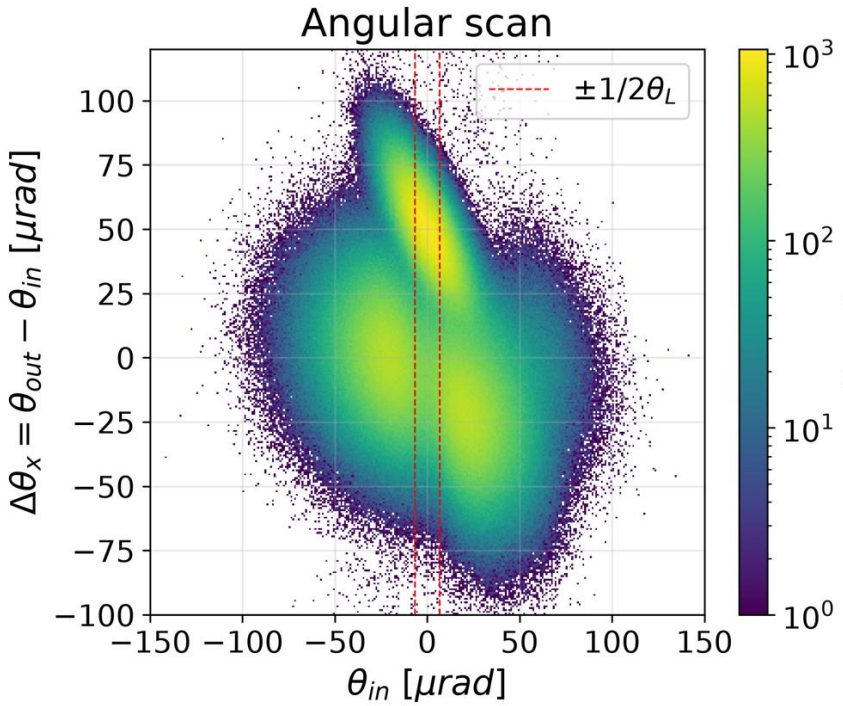


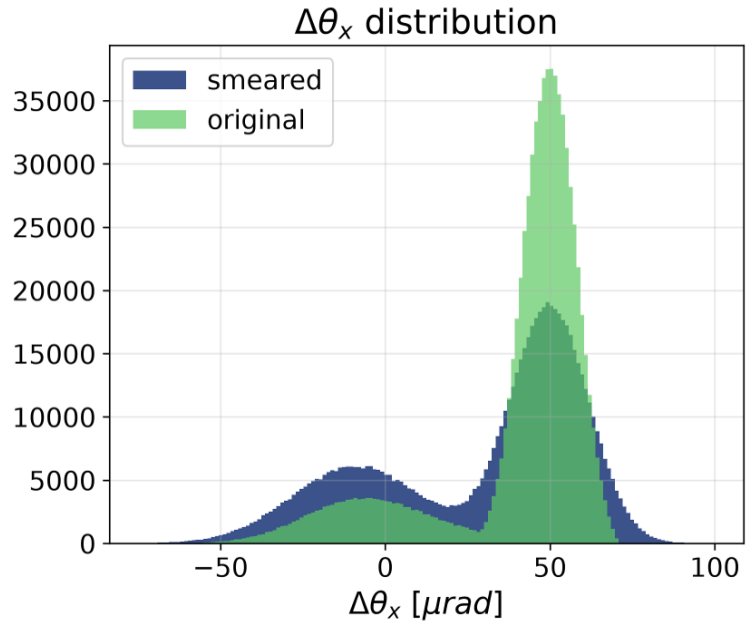


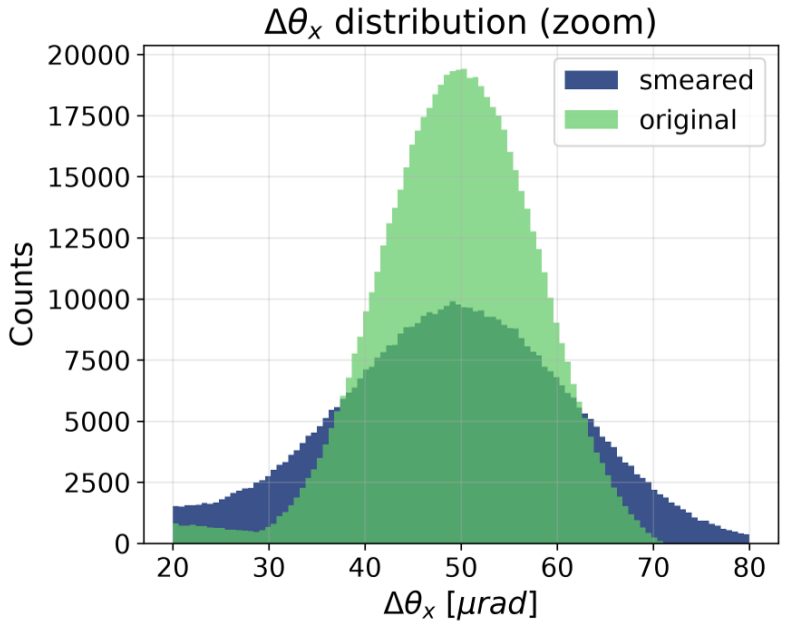
without smear $\varepsilon_{ch} = (79 \pm 1)\%$

with smear $\varepsilon_{ch} = (62.6 \pm 0.5)\%$

SHORT CRYSTAL: XCOLL



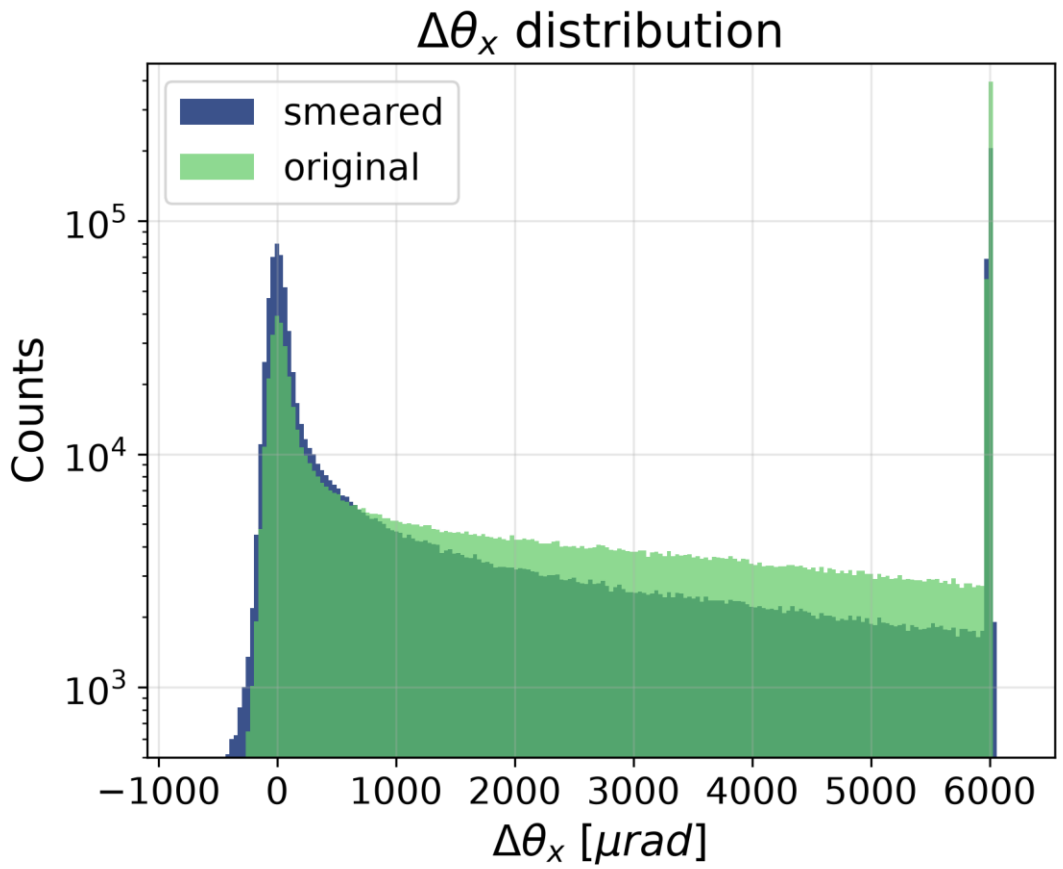


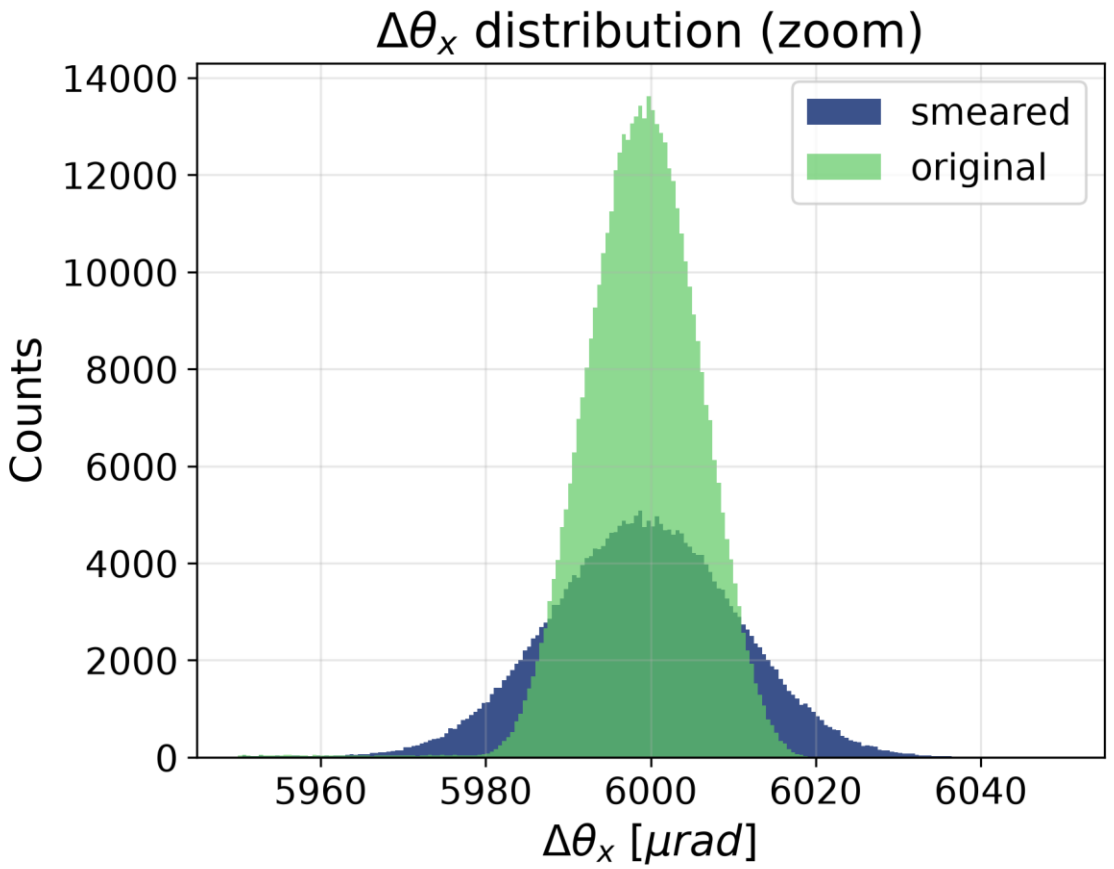


without smear $\varepsilon_{ch} = (75 \pm 1)\%$

with smear $\varepsilon_{ch} = (64.8 \pm 0.2)\%$

LONG CRYSTAL: BDSIM

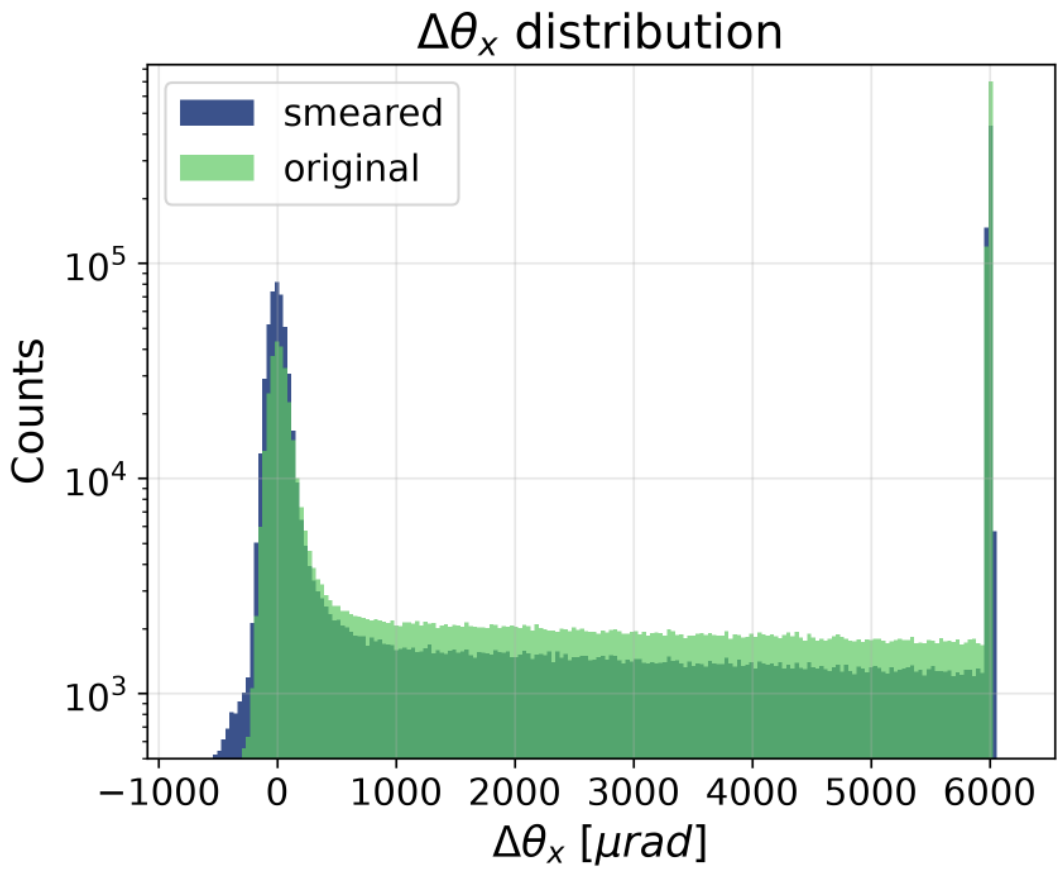


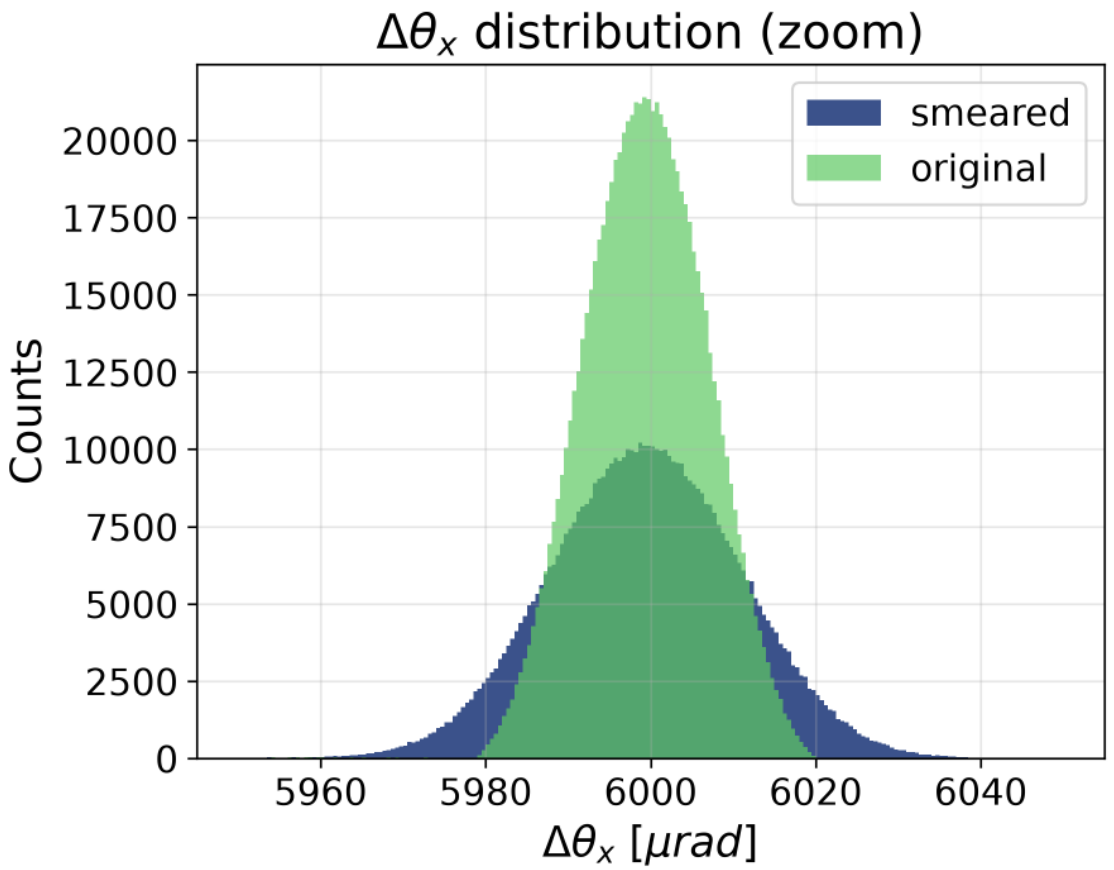


without smear $\varepsilon_{ch} = (30.3 \pm 0.6)\%$

with smear $\varepsilon_{ch} = (20.2 \pm 0.1)\%$

LONG CRYSTAL: XCOLL

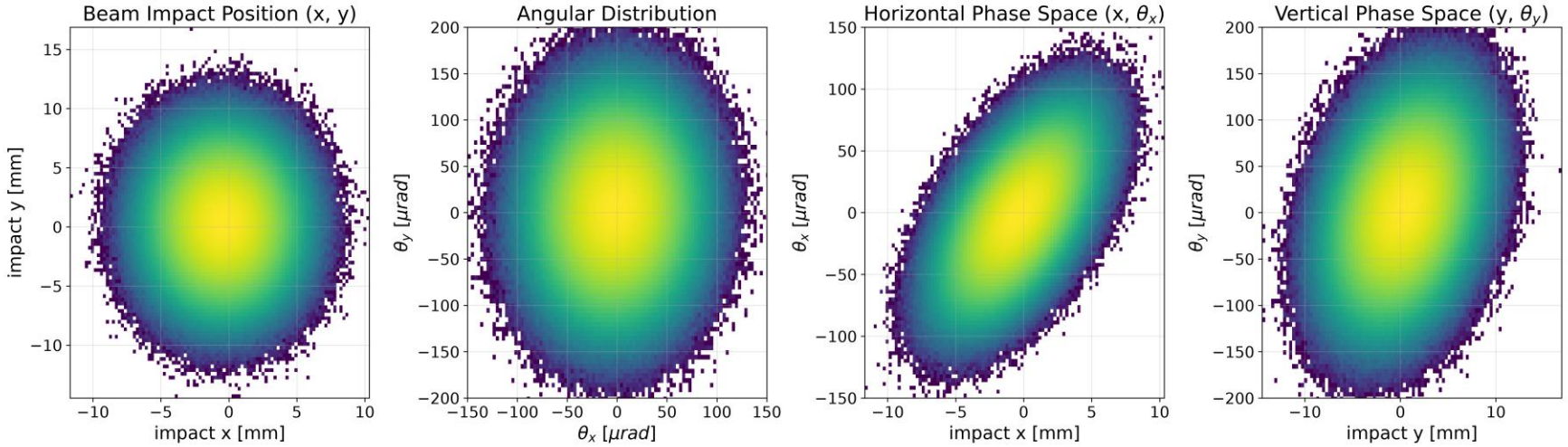


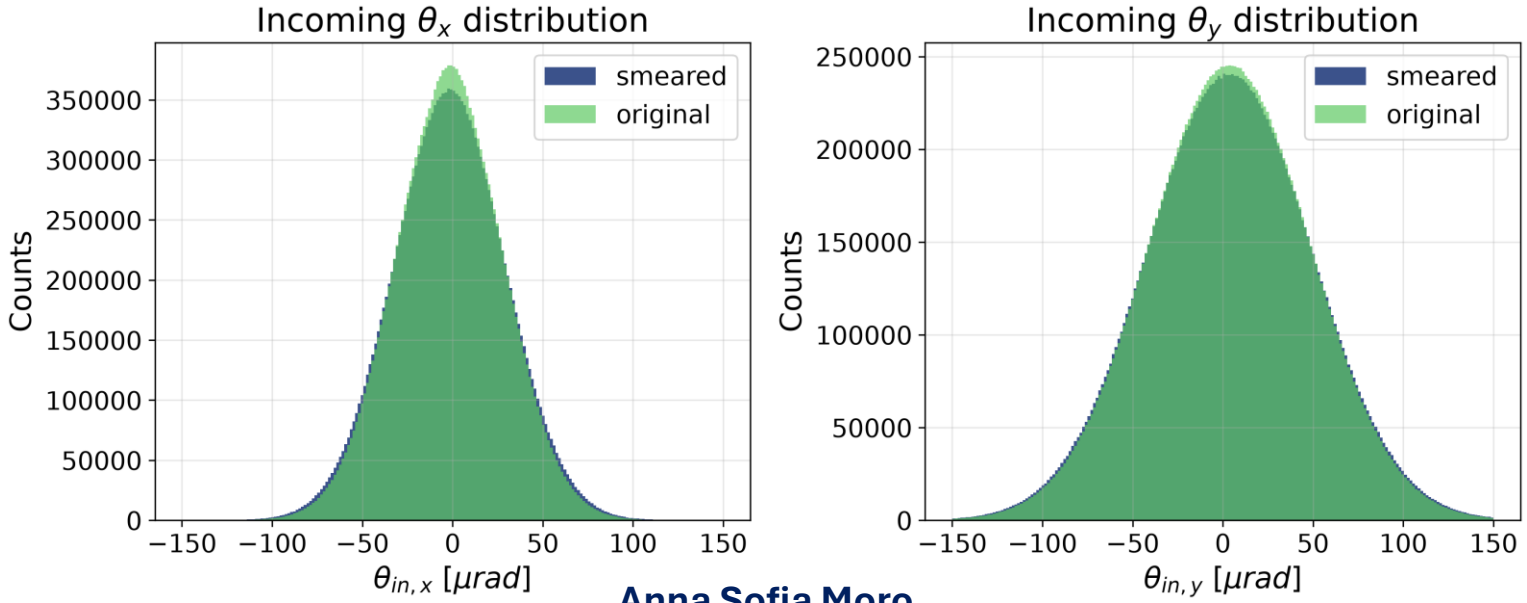


without smear $\varepsilon_{ch} = (49.0 \pm 0.9)\%$

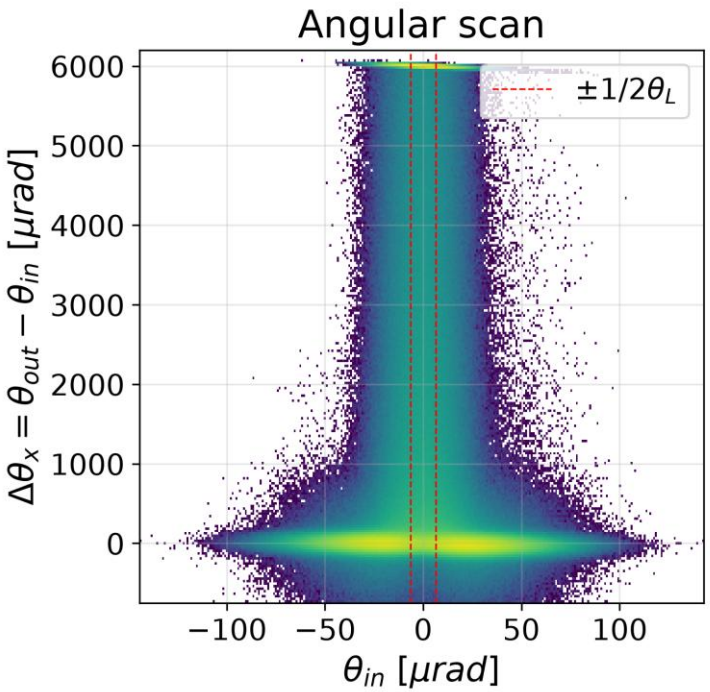
with smear $\varepsilon_{ch} = (41.2 \pm 0.1)\%$

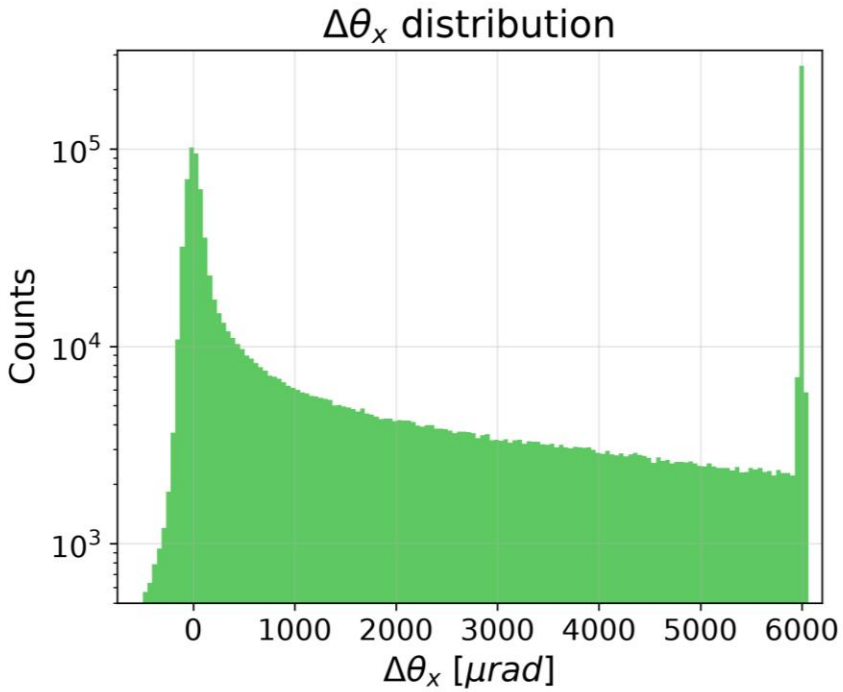
SIMULATIONS

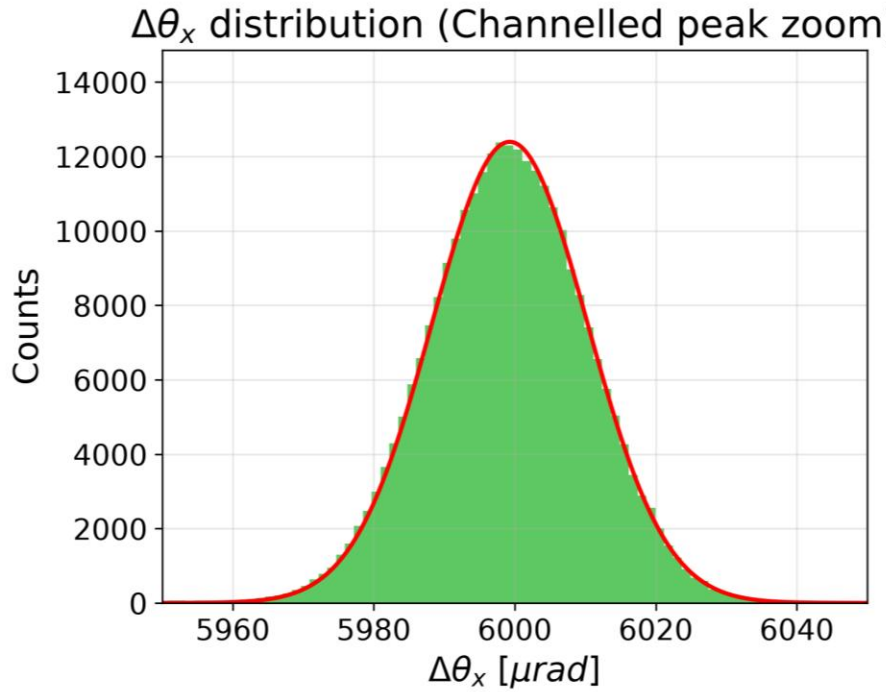




BDSIM – LONG CRYSTAL







$$\theta_b = 5999.3 \pm 0.1\ \mu\text{rad}$$
$$\sigma = 11.0 \pm 0.1\ \mu\text{rad}$$

SHORT CRYSTAL

